The Shape Under the Ice

Student copy (project or hand out)

1 We have better maps of the surface of Mars than of the rock beneath the East Antarctic ice. That is the ordinary condition of the place. Kilometers of ice sit on top of a continent almost nobody has seen, and what is known of the bedrock has been pieced together from radar, gravity readings, and the slow inference of signals the ice does not give up easily.

2 Now a team writing in the journal Nature Geoscience has put a name to a piece of it. They call it the East Antarctic Fan-shaped Basin Province. It is, on their reading, a single structure: roughly thirty subglacial basins that had been catalogued one at a time, set side by side, turn out to splay from one point like the ribs of a fan. The point sits within four degrees of the South Pole. The fan opens out from there toward the coast.

3 The basins are not minor features. The province gathers in the Wilkes and Aurora basins and the trough that holds Lake Vostok, the largest body of liquid water known to lie under any ice sheet on Earth. These were familiar landmarks. What the study argues is that they were never separate landmarks at all. They are fingers of the same hand.

4 The shape has a cause. The lead author, Egidio Armadillo of the University of Genoa, with colleagues including Guy Paxman at Durham, read it as the record of distributed rotational extension: crust that stretched outward from a center, slowly, in more than one episode, as the old supercontinent of Gondwana came apart and Antarctica and Australia went their separate ways. The triangular basins are the gaps that opening left behind.

5 None of this is dead history. The bed under an ice sheet is not a floor. It is a set of channels and dams. Where the rock dips, ice pools and slides; where it rises, ice slows. So a fan drawn into the crust in the deep geological past still has a say in where the ice of East Antarctica travels now, and in which parts of it are most exposed as the climate warms. That last part is where the work goes quiet. The map is new, the consequences are not yet drawn, and the authors are careful to say which is which.

AP-style multiple choice:

1. The opening sentence comparing maps of Mars to maps beneath the Antarctic ice chiefly functions to

A) argue that planetary science is overfunded relative to polar research
B) establish, through juxtaposition, how little of the continent's bedrock is known
C) introduce a personal anecdote about the writer's fieldwork
D) predict that the ice sheet will soon be fully mapped

2. The short sentences such as “The shape has a cause.” set against the longer technical sentences primarily create an effect of

A) hesitation about whether the findings are sound

B) a measured alternation in syntax that carries the reader from plain fact to its complicated explanation

C) impatience with a reader who is slow to understand

D) an emotional plea to protect the ice sheet

Discussion questions:

1. The writer never uses a word like “astonishing” or “breathtaking,” yet the discovery reads as remarkable. How is that effect produced without those words?

2. Compare the simile in paragraph 2 with the metaphor in paragraph 3. What does each comparison do that the other does not?

Writing prompt (homework or extended in-class, Q3 argument):

The article devotes great effort to mapping a landscape no one can see or use directly. Defend, challenge, or qualify the claim that there is real value in studying things we cannot yet put to use. Use your own knowledge, observation, or reading.